Course 4: Project Execution: Running the Project

Module 1-Introduction to project execution

Tracking: A method of following the progress of project activities.

Deviation: Anything that alters your original course of action. Deviations from the project plan can be positive or negative.

Transparency is essential for accurate decision-making.

Tracking keeps all team members and stakeholders in touch with deadlines and goals.

Tracking is also crucial for recognizing risks and issues that can derail your progress.

Tracking helps build confidence that the project is set to be delivered on time, in scope, and within budget.

Commonly tracked items

- Project schedule
- Status of action items, key tasks and activities
- Progress toward milestones
- Costs
- Key decisions, changes, dependencies and risks to the project

GANTT chart

- Useful for staying on schedule
- Useful for projects with many dependencies, tasks, activities, or milestones
- Useful for larger project teams

Roadmap

- Useful for high-level tracking of large milestones
- Useful for illustrating how a project should evolve over time

Burndown chart

- Useful for projects that require a granular, broken-down look at each task
- Useful for projects where finishing on time is the top priority

Gantt charts

The Gantt chart is one of the most popular tracking methods and can be used for all types of projects. Gantt charts typically live in your project plan and are updated as the project progresses.

Gantt charts are useful for:

- Helping a team stay on schedule
- Projects with lots of tasks, dependencies, and milestones
- Projects with large teams, because ownership and responsibilities are explicitly laid out visually

Roadmaps

Roadmaps are another common tracking method. Like Gantt charts, Roadmaps also track both individual and project progress toward milestones. However, Roadmaps are best suited for tracking big milestones in your project.

Roadmaps are useful for:

- High-level tracking of large milestones. Roadmaps outline the project as a whole and provide an overall snapshot of key points—just like an actual roadmap contains points of interest and mile markers.
- Illustrating to your team or key stakeholders how a project should evolve over time

Burndown charts

Burndown charts are typically used by Agile Scrum teams. Burndown charts reveal how quickly your team is working by displaying how much work is left and how much time remains to complete the work. The main uses of a Burndown chart are to keep the project team on top of targeted completion dates and make them aware of scope creep if it occurs. The chart should be displayed so everyone can see it and needs to be updated regularly in order to be effective.

Burndown charts are useful for:

- Projects that require a detailed review of tasks
- Projects where finishing on time is the top priority

Key takeaways

To recap, project status reports are a powerful tool to:

- Improve and simplify communication across the team.
- Keep everyone, including key stakeholders, informed.
- Request more resources and support (if needed).
- Create structure and transparency by recording the project status in a centralized place.

Examples of project risk

- A contractor misses a deadline
- A new tool leads to a communication breakdown
- Workload increases due to the implementation of an unforeseen policy

Change: Anything that alters or impacts the tasks, structures, or processes within a project

Types of changes

- New or changing dependencies
- Changing priorities
- Capacity and people
- Limitations on budget or resources
- Scope creep
- Force majeure

Dependencies: The links that connect one project task to another and are often the greatest source of risk to a project

Internal dependencies: The relationship between two tasks within the same project

External dependencies: Tasks that are reliant on outside factors, like regulatory agencies or other projects

Mandatory dependencies: Tasks that are legally or contractually required

Discretionary dependencies: Tasks that could occur on their own, but the team chose to make them reliant on one another

Dependency management: The process of managing interrelated tasks and resources within the project to ensure the overall project is completed successfully, on time, and in budget.

- Proper identification
- Recording dependencies
- Continuous monitoring and control
- Efficient communication

Risk register: A table or chart that contains your list of risks and dependencies

Risk management: The process of identifying potential risks and issues which could impact a project, and evaluating and applying steps to address the effects of the identified risks and issues

If your dependencies are met on time, your team is less likely to fall behind schedule. If your scope is tightly managed, you're less likely to incur changes to your budget or be forced to extend your timeline. A risk register is a table or chart that contains your team's list of risks.

Risk exposure: A way to measure the potential future loss resulting from a specific activity or event

The ROAM Framework

1. **Resolved:** Consider this risk to be addressed. It's no longer a problem.
2. **Owned:** Assign a team member ownership of the risk and monitor the risk through to completion.
3. **Accepted:** Understand and accept the risk for what it is, because it can't be resolved.
4. **Mitigated:** Formulate a plan to eradicate the risk.

Escalation: The process of enlisting the help of higher level project leadership or management to remove an obstacle, clarify or reinforce priorities, and validate next steps

Escalations

- Act as checks and balances
- Generate speedy decision making
- Reduce frustrations
- Encourage participation

Before starting work on a project, the project manager, the team, and the project sponsor should establish escalation standards and practices. A project manager should escalate an issue at the first sign of critical problems in a project. Issues that warrant escalation can:

- Cause a delay on a major project milestone
- Cause budget overruns

- Result in the loss of a customer
- Push back the estimated project completion date

Trench wars occur when two peers or groups can't seem to come to an agreement, and neither party is willing to give in.

A bad compromise occurs when two parties settle on a so-called solution but the end product still suffers.

When communicating a small change that will affect an individual, it's a good idea to send an email. When there's a big change within your project that impacts more than one person and is likely to change the budget, deadline, or scope of the project, you'll want to have a team meeting.

Timeout: Taking a moment away from the project in order to take a breath, regroup, and adjust the game plan

Retrospective: A meeting focused on identifying the contributing causes of an incident or pattern of incidents without blaming any individual

Effective escalation emails:

- Maintain a friendly tone
- State your connection to the project
- Explain the problem
- Explain the consequences
- Make a request

Module 2-Quality management and continuous improvement

Quality: When you fulfill the outlined requirements for the deliverable and meet or exceed the needs or expectations of your customers

Quality management components

- Quality standards
- Quality planning
- Quality assurance
- Quality control

Quality standards: Provide requirements, specifications, or guidelines that can be used to ensure that products, processes, or services are fit for achieving the desired outcome

Quality planning: The actions of the project manager or the team to establish a process for identifying and determining exactly which standards of quality are relevant to the project as a whole. Ask yourself:

- What outcome do my customers want?
- What does quality look like for them?
- How can I meet their expectations?
- How will I determine if the quality measure will lead to project success?

Quality assurance (QA): Evaluating if your project is moving towards delivering a high-quality service or product

Quality control (QC): Involves monitoring project results and delivery to determine if they are meeting desired results or not

Ask open-ended questions and actively listen to understand the customers' current state versus their desired state. Set clear expectations about when you'll communicate certain things to customers.

Feedback surveys: A survey in which users provide feedback on features of your product that they like or dislike

UAT agenda steps

- Welcome users and thank them for participating
- Present your product to them
- Start UAT test cases
- Walk users through a demonstration
- Identify edge cases
- Recap findings, identify issues, prioritize which issues should be addressed first

Edge cases: Rare outliers that typically pertain to software based projects. They deal with the extreme maximums and minimums of parameters.

Accessibility Guidelines

- Offer to provide accommodations
- Examine the space with an accessibility lens

- Check that the systems you are using are fully accessible
- Make accessibility part of the conversation from the beginning

Think about accessibility early and often, and encourage others on your project team to do so, too.

Continuous improvement: An ongoing effort to improve products or services. Continuous improvement begins with recognizing when processes and tasks need to be Created, Eliminated, or Improved.

Process improvement: The practice of identifying, analyzing, and improving existing processes to enhance the performance of your team and develop best practices, or to optimize consumer experiences

Control: An experiment or observation designed to minimize the effects of variables

Data-driven improvement frameworks

Techniques used to make decisions using actual data: **Define, Measure, Analyze, Improve, Control**

PDCA Cycle

A four-step process that focuses on identifying a problem, fixing that issue, assessing whether the fix was successful, and fine-tuning the final fix:

1. **Plan:** Identify the issue and root cause, and brainstorm solutions to the problem.
2. **Do:** Fix the problem.
3. **Check:** Compare your results to the goal to find out if the problem is fixed.
4. **Act:** Fine-tune the fix to ensure continuous improvement.

Project: One single-focused endeavor

Program: A collection of projects

Portfolio: A collection of projects and programs across the whole organization

Retrospective: A workshop or meeting that gives project teams time to reflect on a project

Three main purposes of retrospectives:

- Encourage team building
- Facilitate improved collaboration
- Promote positive changes

The emphasis in retrospectives is on continuous improvement and change, instead of recycling old and potentially flawed processes.

Reasons to hold a retrospective

- Missed deadlines or expectations
- Miscommunications between stakeholders
- Reached the end of a sprint
- Product launches and landings
- Record key lessons that other people can learn from

The way you choose to structure your retrospective will depend on your team and workplace.

Retrospective best practices

- Ensure discussion is blameless
- Reflect on positive aspects of the project as well as the negatives

The retrospective should be considered a positive experience, where team members feel comfortable sharing their feedback.

Lessons Learned Template

- Things that went well:
- Things that need improvement:
- Where we got lucky:

Module 3-Data-informed decision-making

Data: A collection of facts or information

Benefits of using data

- Make better decisions
- Solve problems
- Understand performance
- Improve processes
- Understand your users

Metric: A quantifiable measurement that is used to track and assess a business objective

Types of project metrics

- **Productivity metrics:** Metrics that allow you to track the effectiveness and efficiency of your project (Milestones, Tasks, Projections, Duration).
- **Quality metrics:** Metrics that relate to achieving acceptable outcomes (Number of changes, Issues, Cost variance).

Milestone: An important point within the project schedule that indicates progress and usually signifies the completion of a deliverable or phase of the project

Task: An activity that needs to be accomplished within a set period of time

Projection: How you predict an outcome based on the information you have now

Duration: The total time it takes to complete a project from start to finish

Changes: Metrics that show inconsistencies from the initial requirements of the project

Change log: A record of all notable changes on a project

Issue: A known and real problem that may affect the ability to complete a task

Cost variance: The difference between the actual cost and the budgeted cost

Signal: An observable change that helps determine the overall health of the project and identify early signs that something isn't quite right

Data Priorities

- Identify which tasks contribute most to the overall goal
- Prioritize the data or metrics that are most valuable to stakeholders

Stakeholders can look to your project plan for a high-level overview for answers to important questions, determined success criteria, artifacts, and the overall health of your project.

Data ethics: The study and evaluation of moral challenges related to data collection and analysis. This includes generating, recording, curating, processing, sharing, and using data in order to come up with ethical solutions. Businesses apply data ethics practices so they can:

- Comply with regulations
- Show that they are trustworthy
- Ensure fair and reasonable data usage

- Minimize biases
- Develop a positive public perception

Data analysis: The process of collecting and organizing information to help draw conclusions

Quantitative data: Statistical and numerical facts

Qualitative data: Subjective qualities that can't be measured with numerical data

Conducting a data analysis is an essential process for understanding a business' needs and challenges and determining effective solutions. These six foundational steps—ask, prepare, process, analyze, share, and act—will help set you up for success!

Storytelling: The process of turning facts into narrative to communicate something to your audience

Six Steps of Data Storytelling

1. Define your audience
2. Collect the data
3. Filter and analyze the data
4. Choose a visual representation
5. Shape the story
6. Gather your feedback

Ask yourself: What are their most urgent concerns? Which key data points influence the story and project outcome? What would my audience like to know about the project? Visualizations are a great way to help people remember the information you are presenting and are an essential piece of storytelling.

Data visualization: Is the graphical representation of information to facilitate understanding. It helps to:

- Filter information by focusing the audience on the most important data points and insights
- Condense long ideas and facts into a single image or representation
- Make sense of the information being presented

Dashboard: Type of user interface that provides a snapshot view of your project's progress or performance

KPI: A measurable value or metric that demonstrates how effective an organization is at achieving key objectives

Infographics: Visual representations of information or data intended to present information quickly and clearly

Giving effective presentations

- **Be precise:** Identify the problem you're solving for your audience. Remove any content that dilutes your message.
- **Be flexible:** Consider the approach you'd take if you had to shorten your presentation unexpectedly. Practice to avoid mistakes that could distract from your message. Identify and come up with answers to potential audience questions. Imagine and prepare for possible objections.
- **Be memorable:** Use stories or repetition to help your audience remember information. Beware of your body language: Posture, Tone of voice, Pace, Eye contact, Warm and friendly facial expressions, Confidence.

Presentation Accessibility Checklist

1. Create clear, simple slides
2. Add alt text for images, drawings, or diagrams
3. Use text for critical information
4. Provide captions for all audio or video recordings
5. For contrast and text size, more is better
6. Share content in advance

Contrast ratio: The difference between text and its background color

Module 4-Leadership and influencing skills

Great project managers: Support people to do their best work, enable people to build things they're proud of, define teamwork, and explain why teamwork is important for successful project management.

Team: A group of people who plan, solve problems, make decisions, and review progress in service of a specific project or objective

Work groups: People in an organization who work toward a common goal. Work groups are more likely to be coordinated, controlled, or assigned by a single person or entity. Project managers develop and lead effective teams by fostering a culture of teamwork.

Teamwork: An effective, collaborative way of working in which each person is committed to and heading towards a shared goal

Why teamwork matters in project management

- Fosters creativity
- Encourages accountability
- Helps you get stuff done

Five factors that impact team effectiveness

1. **Psychological safety:** An individual's perception of the consequences of taking an interpersonal risk. Be direct and kind.
2. **Dependability:** Team members are reliable and complete their work on time.
3. **Structure and clarity:** An individual's understanding of job expectations, knowledge of how to meet those expectations, and the consequences of their performance.
4. **Meaning:** Finding a sense of purpose either in the work itself or in the results of that work.
5. **Impact:** The belief that the results of one's work matters and creates change.

Role of the Project Manager

- Create systems that turn chaos into order
- Communicate and listen
- Promote trust and psychological safety
- Demonstrate empathy and create motivation (Being present, Listening, Asking questions)
- Delegate responsibility and prioritize
- Celebrate team success

Create a team atmosphere where different opinions are welcome and members remain respectful of one another. Providing and accepting feedback is a healthy part of project management and usually makes for a better project outcome.

Bruce Tuckman's stages of team development

- **The Forming stage:** Team gets to know one another. Project manager should clarify project goals, roles, and context about project.

- **The Storming stage:** Frustrations emerge. Project manager should focus on conflict resolution, listen as the team addresses problems to solve, and share insights on how the team might better function as a unit.
- **The Norming stage:** Conflict is mostly resolved and team is working together. Project manager should codify the team norms, ensure that the team is aware of those norms, and reinforce them when needed.
- **The Performing stage:** Team works together seamlessly. Project manager should focus on delegating, motivating, and providing feedback to keep up the team's momentum.
- **The Adjourning stage:** Project wraps up, team disbands. Celebrate final milestones and successes.

Team dynamics: The forces, both conscious and unconscious, that impact team behavior and performance. Managing team dynamics is important because teams have individuals with different skill sets, varying degrees of autonomy, and competing priorities. It creates a collaborative environment and helps you understand how to motivate your team.

Ethical leadership: A form of leadership that promotes and values honesty, justice, respect, community and integrity. Promote it by defining and aligning values within your team and demonstrating how adhering to those values benefits the mission of the organization. Create forums where employees can raise viewpoints, feel heard, and receive follow-ups on their concerns.

Inclusive leadership: A form of leadership where everyone's unique identity, background, and experiences are respected, valued, and integrated into how the team operates. It aims to put what we've heard into action.

Diversity: The set of differences each of us possesses, whether visible or invisible, that give us each a unique perspective on the world and our work

Inclusion: What the team does with the diversity of thought and perspective

Project managers lead inclusivity by:

1. **Fosters a culture of respect:** Modeling the values of your organization, taking appropriate action, creating a comfortable environment for communication, recognizing team contributions.
2. **Creating equal opportunity to succeed:** Regular communication, accessible documentation, regular check-ins with the team.
3. **Inviting and integrating diverse perspectives:** Create a sense of psychological safety, invite teammates to share thoughts, ideas, and concerns.

Influencing: The ability to alter another person's thinking or behaviors

Conger's Four Steps to Influence

1. **Establish credibility:** Make the case for why your audience should listen to you. Drawing from expertise and relationships.
2. **Frame for common ground:** Make the case for how your idea can benefit your audience by understanding their values.
3. **Provide evidence:** Make your case through hard data combined with vivid language, examples, and persuasive storytelling.
4. **Connect emotionally:** Demonstrate to your audience that you're emotionally committed to your idea and match their emotional state.

Common influencing mistakes

- Approaching the audience aggressively
- Resisting compromise
- Failing to establish credibility, frame for common ground, or connect emotionally
- Assuming agreements can be worked out in a single conversation

Sources of Power

- **Organizational sources of power:** Role (position), Information (access/control), Network (professional/personal connections), Reputation (perception).
- **Personal sources of power:** Knowledge (expertise/unique skills), Expressiveness (communication ability), History (personal history with someone), Character (view of personal qualities).

As you build your career, try to identify your own superpowers.

Module 5-Effective project communication

Project document responsibilities

- How documents get used
- Who has access to documents
- How often documents get updated

You'll need to communicate certain information to your team multiple times and in various ways. Keep in mind modes of processing: some people learn by listening, some by watching, and some by doing.

Common communication tools: Messaging, Virtual meetings, Work management and collaboration tools.

Email best practices

- Carefully select who you're sending emails to and why
- Make sure the Subject clearly previews what the message is about
- Keep messages short and stay on topic
- Attach or link large amounts of information separately
- Clearly state action items
- Use correct grammar and spelling
- Write in an appropriate tone (Be friendly, motivating, clear, and specific)

Effective meetings characteristics

- **Structured:** Start and end on time, carefully selected attendees, prioritized topics, designated notetaker. Timeboxing means setting a time limit.
- **Intentional:** Clearly stated purpose and expectations. Everyone understands why they're meeting. Send any pre-reading materials in advance.
- **Collaborative:** People working together on an activity. Let everyone know that they're encouraged to respond verbally, through chat, or notes.
- **Inclusive:** The practice of including people who might otherwise be excluded or marginalized. Leave space for quiet participants and ensure meetings are accessible.

Creating an Inclusive Meeting Environment

Inclusive meetings foster a sense of connection and encourage participants to contribute their best work. Project managers should establish consistent procedures like initial check-ins, open-ended questions, and model empathy and professionalism to create emotional safety.

Actively solicit input from all participants, especially those who may feel intimidated or underrepresented. Encourage differing viewpoints and follow up after meetings to gather additional thoughts, using diverse images in materials to support inclusion.

Ensure meetings accommodate people with disabilities by using large fonts, high-contrast colors, and accessible electronic formats for materials. Provide accommodations such as closed captions, accessible seating, clear communication, and simple layouts to support participants with visual, hearing, mobility, or neurological challenges.

Meeting Timelines

- **Before the Meeting:** Prepare and share a clear agenda stating the meeting's purpose and goals. Invite only necessary participants who can contribute and clarify their roles.
- **During the Meeting:** Start by stating meeting goals and stick to the agenda to maintain focus. Encourage active participation, practice active listening, and ensure everyone, including remote attendees, can contribute.
- **After the Meeting:** Recap key decisions, action items, and timelines, then share with participants. Schedule follow-ups if needed and reassess the frequency of recurring meetings to optimize productivity.

Types of project management meetings

- **Project kick-off:** The official beginning of a project and serves as a way to align the team's understanding of the project goals with actual plans and procedures.
- **Status updates:** Regular team meetings where the primary goal is to align the team on updates, progress, challenges, and next steps. Covers task updates, schedule status, and budget status. Discusses current or anticipated issues (changes, risks, resource issues, vendor issues, action items). Update frequency depends on project complexity, number of team members, and the level of information required by the sponsor or clients.
- **Stakeholder reviews:** The goal is to get buy-in and support. Topics include presenting a project update, seeking/listening to feedback, and making a decision or resolving a major issue.
- **Project reviews / Retrospective meetings:** Topics include lessons learned (what worked, what to improve) and celebration.

Module 6-Closing a project

Project closing: The process performed to formally complete the project, the current phase, and contractual obligations. It aims to assure all work is done, agreed-upon project management processes are executed, and formal recognition and agreement that the project is done is received from key stakeholders.

The never-ending project exists when the project deliverables and tasks cannot be completed.

The abandoned project exists when inadequate handoff or transition on the project deliverables occurs.

Consequences of Skipping Closure Steps

- Incomplete work can lead to costly rework and missed deadlines, as seen in the Tilly's Toys case with missing safety disclaimers.
- Failure to complete agreed-upon processes, such as signing NDAs, can result in legal risks and breaches of contract.
- Lack of formal project closure can cause misunderstandings with stakeholders and damage future business relationships.
- Skipping closure steps can delay product launches, cause financial losses, and harm team and organizational credibility. Project managers should plan and execute the closing phase with the same rigor as other project phases.

Closing Checklists

A small, closing process can happen at the end of each milestone, or a formal, larger comprehensive closing phase at the very end only.

Steps to conducting a closing process after each phase or milestone:

- Refer to prior documentation: Statement of Work (SoW), Request for Proposal (RFP), Risk register, RACI chart.
- Put together closing documentation.
- Conduct administrative closure of the procurement process.
- Make sure all stakeholders are aware that a phase, or project, is ending.
- Execute necessary follow-up work.

Steps to conducting a closing process at the end of your project:

- Provide the necessary training, tools, documentation, and capability to use your product.
- Ensure that the project has satisfied its goals and desired outcomes.
- Document acceptance from all stakeholders.
- Conduct a formal retrospective.
- Disband and thank the project team.
- Review all contracts and documentation.

Impact reporting: Presentation that is given at the end of a project for key stakeholders.

- **Highlight Key Performance Areas:** The impact report should show how the project met its goals, objectives, budget, schedules, and key performance indicators (KPIs) set at the start. It should answer what problem the project aimed to solve and how it was solved, showcasing the value brought to the organization.

- **Use Metrics to Showcase Results:** Use facts, statistics, and relevant metrics such as schedule improvement, revenue growth, ROI, user counts, cost savings, and customer satisfaction to demonstrate success. Collect and track data throughout the project and complement metrics with visuals tied to the project's goals.
- **Prepare an Effective Presentation:** Be concise, organize information clearly, and avoid excessive technical jargon to ensure stakeholder understanding. Use visuals like charts, graphs, images, and icons; engage stakeholders with stories, videos, or interactive elements; and discuss lessons learned and areas for improvement.

Retrospective: A meeting aimed to discuss successes, failures and possible future improvements on the project. Three benefits of retrospectives include encouraging team building, facilitating improved collaboration, and promoting positive changes.

Project closeout report

- It is a blueprint to document what the team did, how they did it, and what they delivered.
- It provides an evaluation of the quality of work.
- It evaluates the project's performance with respect to budget and schedule.

Project Closeout Report Components

- Includes essential details such as date, status, project team members, and project duration.
- Contains an Executive Summary outlining project scope, deliverables, and issues encountered.
- **Key Sections:** Key Accomplishments (highlight tasks, milestones, goals, and results achieved) and Lessons Learned (cover both successes and problems, including solutions implemented).
- **Ongoing and Future Considerations:** Open Items (areas needing improvement and ongoing maintenance) and Next Steps and Future Considerations (suggest actions for future project managers).
- **Additional Elements:** Project Timeline (shows start and end dates for tasks/milestones) and Resources and Project Archive (link to important project documents for reference).